

OOP Practice Problems

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Class, Object

Shapes

Point

- Write a class named ***Point*** having two member variables **x(int)** and **y(int)**.
- Write a method ***distance*** inside the ***Point*** class to calculate the distance between two points. The function looks like this -

```
public double distance(Point p){ // codes }
```

Formula for distance between two points **(x1, y1)** and **(x2, y2)** -

$$\sqrt{(x1 - x2)^2 + y1 - y2)^2} \quad (1)$$

Circle

- Write another class named ***Circle*** having two instance variables **center(Point)** and **radius(double)**.
- Write a method inside ***Circle*** class that checks if a point resides inside a circle. The function looks like this -

```
public boolean pointInsideCircle(Point p){ // codes }
```

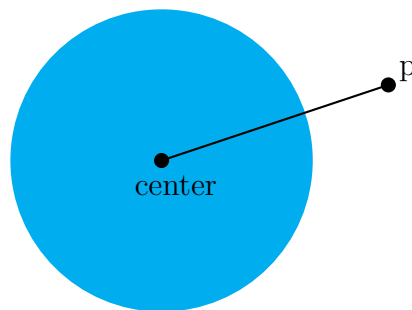


Figure 1: Check if the point p is inside the blue circle

- Write a method named *circleInsideCircle* inside *Circle* class that checks if a circle resides into another circle. The function looks like this

-

```
public boolean circleInsideCircle(Circle C){ // codes }
```

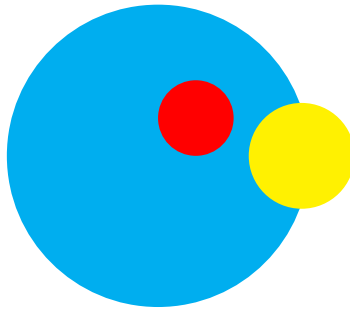


Figure 2: Red colored circle is inside of the Blue colored circle, but the yellow one is not inside

Rectangle

- Write a class *Rentangle* having two instance variables *bottomLeft(Point)* and *topRight(Point)*.

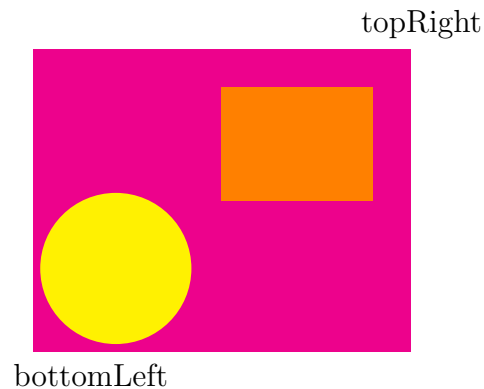


Figure 3: Check for the yello wcircle and orange rectangle if they are inside of the big traigle

- Write a method named *circleInsideRectangle* inside *Rectangle* class that checks if a circle resides into a Rectangle. The function looks like this -

```
public boolean circleInsideRectangle(Circle R){ // codes }
```

- Write a method named *RectangleInsideRectangle* inside *Rectangle* class that checks if a Rectangle resides into another Rectangle. The function looks like this -

```
public boolean RectangleInsideRectangle(Rectangle R){ // codes }
```

BangladeshiFlag

- Write a class *BangladeshiFlag* having two instance variables *R(Rectangle)* and *C(Circle)*.
- Write a method named *isBalanced* inside *BangladeshiFlag* class that checks if a flag is well-balanced. You need to check if the circle resides exactly in the middle of the rectangle without touching the rectangle. The function looks like this -

```
public boolean isBalanced(){ // codes }
```

